APNet: A Novel Antiperturbation Network for Robust Hyperspectral Image Classification Against Adversarial Attacks

Lin Zhao, *Member, IEEE*, Youlin Zhang, Chengzhong Shi, Minhui Zhao, Jianhui Wu, and Wen Li

***Abstract*— Deep learning (DL) methods have achieved impressive performance in hyperspectral image (HSI) classification but are susceptible to adversarial attacks, which can lead to significant accuracy degradation. Although there has been encouraging progress in improving model robustness for HSI classification, the existing approaches primarily concentrate on capturing global pixel-level dependencies while overlooking the intrinsic superpixel priors of HSI—namely, spatial smoothness and spectral correlations within spatially coherent regions. In addition, these methods often lack effective noise suppression mechanisms. To address these challenges, we introduce a novel antiperturbation network, APNet, designed for robust HSI classification against adversarial attacks. APNet incorporates a noise suppression module that includes a U-shaped unit (UsU) and a feature cascade unit (FCU) to extract clear pixel-level features at multiple scales. These features are then combined with superpixel priors, which serve as robust tokens for a Transformer encoder to capture global structural features. By fusing denoised pixel-level features with coarse-grained superpixel-level features, APNet significantly enhances the robustness of feature representations and the network's intrinsic resistance to adversarial attacks. Extensive experiments on three HSI benchmark datasets show that APNet outperforms the existing state-of-the-art techniques against across various attacks and perturbation intensities. In particular, APNet maintains stable classification performance even under high attack intensities.**

***Index Terms*— Adversarial attack, adversarial defense, deep learning (DL), hyperspectral image (HSI) classification, superpixel prior.**

I. Introduction

HYPERSPECTRAL image (HSI) provides high spectral resolution data comprising of hundreds of narrow bands. This wealth of spectral information goes far beyond that provided by traditional color or multispectral remote sensing (RS) images. As one of the most important applications of HSI, HSI classification tasks have been widely applied in various RS domains, such as vegetation assessment [1], precision agriculture [2], and environmental monitoring [3].

With the advancement of deep learning (DL) technology, traditional HSI classification methods based on handcrafted features have largely been replaced by DL-based approaches. Among these, convolutional neural networks (CNNs) have achieved notable performance due to their strong generalization ability. However, despite their effectiveness in capturing local information, CNNs have limitations in capturing global dependencies. To address this issue, researchers have integrated CNNs with Transformers to generate representations that combine local details with global dependencies [4], [5], [6], [7]. In addition to these hybrid models, recent innovations, such as SpectralMamba [8] and SpectralGPT [9], have shown great promise in advancing HSI classification. SpectralMamba introduces an efficient state-space model that mitigates spectral variability, while SpectralGPT leverages self-supervised learning to develop a universal RS foundation model, indicating substantial potential in RS applications. As DL-based HSI classification becomes increasingly central to HSI vision applications, there is growing attention to its security implications.

Over the past decade, Szegedy et al. [10] discovered that deep neural networks (DNNs) used for image classification are vulnerable to attacks through carefully crafted input samples. These samples, in the form of small perturbations imperceptible to human vision, can readily deceive well-performing DNNs. Such attacks, explicitly called adversarial attacks by Goodfellow et al. [20], reveal the vulnerability of DL that can be maliciously exploited, with profound social implications. For example, an attacker could cause self-driving vehicles to misinterpret traffic signs or road markings by generating adversarial examples [11], [12].

In recent years, the RS community increasingly recognized the threat posed by adversarial attacks. Czaja et al. [13] introduced the concept of adversarial example to the RS community for the first time, demonstrating that the DL model can be manipulated to make incorrect prediction by adding a small patch in the center of RS image (RSI). Subsequently, work by Chen et al. [14] investigated the adversarial example problem in RS scene classification, attributing model vulnerability both model structure and the RSI dataset.

Received 8 May 2024; revised 31 August 2024; accepted 3 September 2024. Date of publication 10 October 2024; date of current version 18 October 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62473148; in part by the Natural Science Foundation of Hunan Province of China under Grant 2023JJ30284, Grant 2019JJ40104, and Grant 2020JJ4343; in part by the Scientific Research Projection of the Education Department of Hunan Province under Grant 23A0488 and Grant 22C0365; and in part by the Graduate Research and Innovation Project of Hunan Province under Grant CX20190930 and Grant CX20190932. *(Corresponding author: Wen Li.)*

The authors are with Hunan Engineering Technology Research Center for 3D Reconstruction and Intelligent Application, Hunan Institute of Science and Technology, Yueyang 414000, China (e-mail: 446516001@qq.com).

Digital Object Identifier 10.1109/TGRS.2024.3467088

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

To tackle the impact of adversarial attacks on HSI classification model, defense strategies can be broadly divided into two categories: adversarial training and tailored network structure. Although adversarial training is a well-established defense method, it has received relatively little attention in the hyperspectral field due to its requirement for additional training examples and substantial computational resources. For instance, Shi et al. [15] proposed a modified DeepFool algorithm to generate boundary adversarial samples and then integrate these adversarial samples into training set to improve model training. In contrast, tailored network structures have shown great promise in defending against adversarial attacks. Xu et al. [16] introduced a novel self-attention (SA) context network that achieved state-of-the-art performance on adversarial test sets. They also developed a spatial–spectral SA learning network specifically tailored to defend against adversarial attacks on HSI classification [17]. Furthermore, Qi et al. [18] utilized random input masking and self-supervised mechanisms to develop a robust architecture. These tailored network approaches significantly improve model security, with superior performance largely attributed to their ability to capture global contextual information.

Undoubtedly, the aforementioned methods have made significant strides in defending against adversarial attacks on HSI classification networks. However, they primarily focus on capturing spectral information and spatial features while often overlooking the importance of spatial coherence constraints and the specific network design required to mitigate perturbations. As a result, they struggle to effectively eliminate or filter external perturbations at the source, especially in cases involving larger perturbations. Therefore, there remains considerable potential to further explore the intrinsic characteristics of HSI and enhance the robustness of DNNs by fully exploiting these features. As illustrated in Fig. 1, neighboring pixels in the HSI are often spatially close and have similar spectral signatures due to the spatial smoothness of the scene and the spectral correlation between adjacent pixels. Notably, when the perturbations are introduced to the input HSI, the superpixel segmentation algorithm simple linear iterative clustering (SLIC) [19] is less sensitive, while the performance of the network model is seriously impaired. The main reason is that SLIC incorporates spatial coherence constraints, which can help disperse subtle perturbations by aggregating spectral information within spatially coherent regions. On the other hand, DNNs are characterized by a large number of parameters and high nonlinearity, enabling them to learn complex patterns and representations from HSI. However, this complexity also makes them vulnerable to adversarial attacks. As shown in Fig. 1, a 2D-CNN that achieves an overall accuracy (OA) of 96.66% on the original clean image sees its OA drop dramatically to just 29.09% when subjected to an adversarial example.

To address these challenges, we introduce APNet, a novel antiperturbation network designed for robust HSI classification that leverages the intrinsic spatial coherence of superpixels and suppresses adversarial perturbations through a U-shaped CNN architecture. APNet incorporates two key innovations: the full utilization of superpixel priors and the effective suppression of adversarial perturbations. First, a noise suppression module is employed to extract clear pixel-level features. These features are then combined with superpixel priors to generate robust tokens, which are subsequently fed into a Transformer encoder to capture global structural features. Finally, the fusion of these global structural features with pixel-level features results in a comprehensive and effective representation for perturbation rejection. The main contributions of this article are as follows.

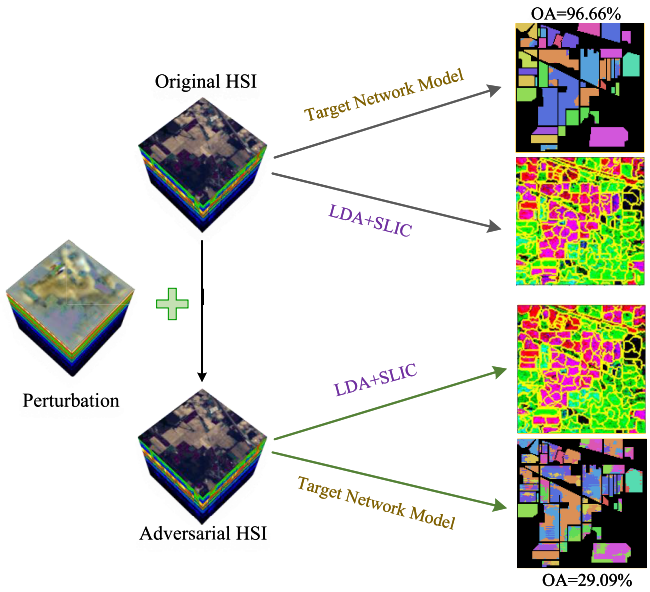


Fig. 1. Schematic of adversarial attacks on DL-based HSI classifier and HSI superpixel segmentation algorithm.

1) We introduce the design concepts of exploiting superpixel priors and mitigating noisy perturbations. APNet employs a U-shaped CNN for perturbation suppression and utilizes Transformers to capture spatial dependencies from superpixel maps. This design allows the network to learn more robust feature representations by effectively integrating both denoised pixel-level and coarse-grained superpixel-level features.
2) To extract clear pixel-level features, we propose a noise suppression module that includes a U-shaped unit (UsU) and a feature cascade unit (FCU). The UsU uses encoder–decoder to effectively attenuate perturbations within the feature map. Subsequently, the FCU progressively extracts spatial and spectral information, refining and enhancing pixel-level feature representations.
3) To further improve adversarial robustness, we integrate superpixel priors with pixel-level features to generate superpixel sequences and introduce a Transformer encoder to capture the global spatial structure and inter-superpixel relationships within these sequences. This significantly strengthens the network's resistance to adversarial attacks, reducing its vulnerability to subtle perturbations in the input space.
4) Extensive experiments on three HSI widely used datasets demonstrate that APNet outperforms the existing state-of-the-art techniques across various attack types and intensities. Notably, APNet maintains stable classifica-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

tion performance even under high attack intensities, highlighting its intrinsic validity and robustness.

The remaining sections of this article are organized as follows. Section II reviews adversarial attacks and defense methods. Section III describes the proposed APNet in detail. Section IV describes the dataset used for the experiments and analyzes the experimental results. Finally, this article is summarized in Section V.

II. Related Works

A. Adversarial Attack Methods

This section will specifically introduce three typical and representative methods for generating adversarial examples.

1) Fast Gradient Sign Method: The fast gradient sign method (FGSM) was introduced by Goodfellow et al. [20] in a paper published in 2015. In their paper, the authors highlighted that the linear nature of DNNs is a key factor contributing to their susceptibility to adversarial attacks. They proposed a simple and rapid approach for generating adversarial examples, known as FGSM. The adversarial example, denoted as x_{adv} and computed using the FGSM method, is defined as follows for an original image x and a target label y_{target}:

$$x_{\text{adv}} = \text{Clip}(x - \varepsilon \cdot \text{sign}(\nabla_x J(x, y_{\text{target}}))) \tag{1}$$

where $\nabla_x J(x, y_{\text{target}})$ represents the gradient of the loss function $J(\cdot)$ concerning the input x and $\text{sign}(\cdot)$ denotes the sign function. The parameter ε signifies the magnitude of the adversarial perturbation, and $\text{Clip}(\cdot)$ is a trimming function employed to confine the adversarial examples within a specified range.

Typically, the success rate is higher when using adversarial examples generated with larger values of ε to attack a target network. Nevertheless, a higher ε value also leads to more pronounced distinctions between the generated adversarial examples and the original image, rendering them more susceptible to detection by the human eye.

Kurakin et al. [21] further proposed the FGSM with a unitized ℓ_∞. This approach has the advantage of producing adversarial examples with more consistent perturbation sizes, as the perturbation is normalized to a unit size. This normalization makes the adversarial examples more controllable, facilitating better comparison and analysis. The specific definitions are as follows:

$$x_{\text{adv}} = \text{Clip}(x - \varepsilon \cdot \frac{\nabla_x J(x, y_{\text{target}})}{\left\|\nabla_x J(x, y_{\text{target}})\right\|_\infty}) \tag{2}$$

2) Basic Iterative Method/Projected Gradient Descent: To generate adversarial examples that are more difficult to discriminate and get a higher success rate, the basic iterative method (BIM) method was proposed by Kurakin et al. [21]. This method further improves the FGSM method by transitioning from a single-step computation for generating perturbations to a multistep iterative process, thereby facilitating the generation of more effective adversarial examples. The formula is provided below

$$\begin{aligned} x_{\text{adv}}^0 &= x \\ x_{\text{adv}}^n &= \text{Clip}(x_{\text{adv}}^{n-1} - \varepsilon \cdot \text{sign}(\nabla_x J(x_{\text{adv}}^{n-1}, y_{\text{target}}))) \end{aligned} \tag{3}$$

where x_{adv}^n is the adversarial example generated during the nth iteration. Based on BIM, Madry et al. [22] proposed the projected gradient descent (PGD) method, which is mainly different from BIM in that a uniform random noise is added to the original image as initialization.

3) Carlini and Wagner's Attack: The Carlini and Wagner's Attack (C&W) attack is an attack algorithm that uses optimization to generate adversarial examples [23], and the goal of the algorithm is to find a minimum perturbation to make the classifier misclassify. The optimization problem takes the following form:

$$\begin{aligned} &\text{minimize} \|\delta\|_p + c \cdot f(x + \delta) \\ &s.t. \quad x + \delta \in [0, 1]^n \end{aligned} \tag{4}$$

where δ is the added perturbation, $f(\cdot)$ denotes the loss function, and c is a hyperparameter that weighs the degree of perturbation and the success rate of the attack. The C&W attack has a higher attack success rate and a smaller amount of perturbation compared with other gradient-based attack algorithms, so it is often used to evaluate the robustness and security of DNNs.

B. Adversarial Defense Methods

The existence of adversarial examples poses a very serious threat to the use of DL methods in numerous secure domains. Currently, the most commonly used defense methods against adversarial examples can be broadly categorized into three main approaches: adversarial example detection, adversarial training, and network structure modification.

1) Adversarial Example Detection: The defense method of adversarial example detection is an effective strategy against adversarial example attacks. The main idea is to determine whether the input sample provided to the DNNs is a clean examples or an adversarial example; if it is a clean example, the sample is directly passed to the network for prediction; conversely, the sample input is rejected.

Xu et al. [24] proposed a novel adversarial example detection method based on feature compression technology. This method involves compressing the original image through feature compression and then comparing the original sample with the feature-compressed version to detect the presence of adversarial examples. This straightforward and efficient strategy can be used in combination with other defense methods to enhance overall robustness. Furthermore, Meng and Chen [25] proposed the MagNet network architecture as a defense against adversarial example attacks. The framework consists of a detector and a reorganizer, where the detector is used to distinguish normal samples from adversarial examples, and the reorganizer is able to reorganize the adversarial examples into normal samples that are similar to the original samples, thus improving the classification accuracy.

2) Adversarial Training: The concept of adversarial training is to enhance the resistance of DNNs to adversarial attacks by including adversarial examples in their training dataset. This approach was first proposed by Goodfellow et al. [20], who utilized the adversarial examples generated by the FGSM method for adversarial training, while Madry et al. [22]

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

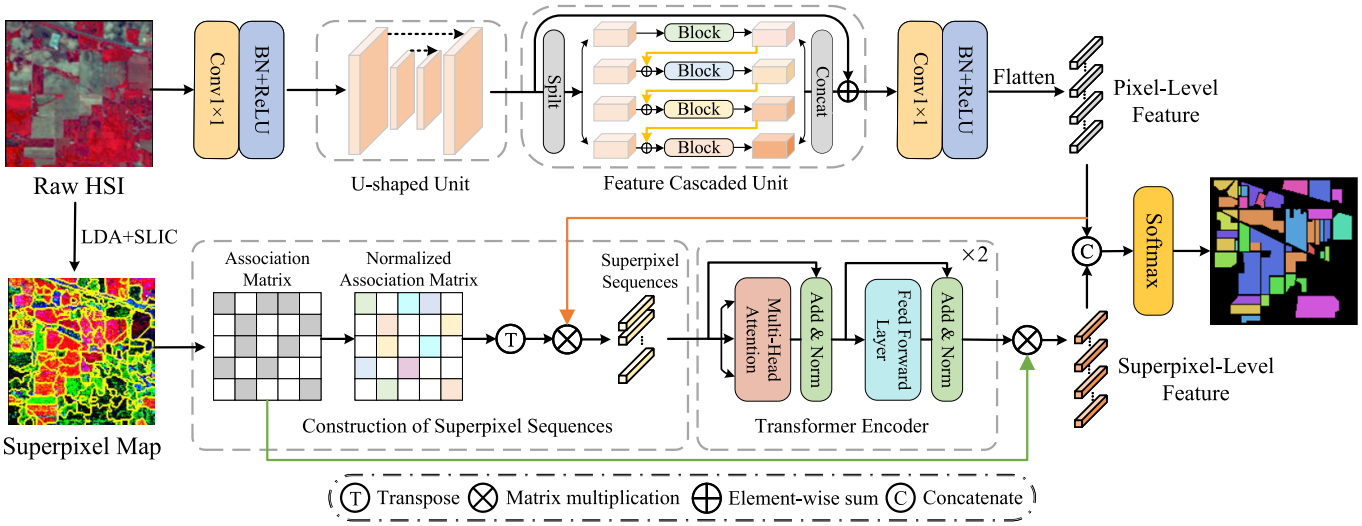


Fig. 2. Architecture of the proposed APNet for HSI classification.

proposed suggesting the use of the PGD method for adversarial training, which resulted in even stronger adversarial robustness for the network.

It is worth noting that while adversarial training improves the adversarial robustness of the network, it comes at the cost of reducing the recognition accuracy of DNNs on clean examples. To strike a balance between adversarial robustness and accuracy after adversarial training, Zhang et al. [26] introduced an adversarial training strategy known as TRADES. In this approach, they designed a novel loss function that combines the natural error and the boundary error and optimized this new loss function through network training, achieving the harmonious unity of DNNs adversarial robustness and accuracy.

3) Network Structure Modification: Adversarial training undeniably enhances a network's resistance to adversarial attacks, but at the cost of decreasing the network's recognition accuracy on clean examples. Moreover, adversarial training does not able to defend against unknown attacks, and the network is still vulnerable to adversarial examples after adversarial training.

To address these limitations, researchers have explored modifying network structures to enhance inherent adversarial robustness. Gu and Rigazio [27] proposed the deep contractive network (DCN), a network designed to enhance network adversarial robustness by incorporating a smoothing penalty that constrains the network's output values. Another approach by Bradshaw et al. [28] involves integrating Gaussian process classifiers into DNNs, resulting in the creation of Gaussian processes hybrid DNNs (GPDNN). GPDNN combines the advantages of both DNNs and gaussian processes to better capture network uncertainty, thereby enabling the network to better withstand adversarial attacks.

III. Proposed Method

The structure of the proposed APNet is shown in Fig. 2. APNet primarily comprises three key components: UsU, FCU, and Transformer encoder. First, we designed UsU to extract the spatial features of HSI, and with cross-layer fusion, UsU can effectively attenuate the perturbations embedded in the feature map to improve the quality of pixel-level features. Second, we design FCU to enhance the pixel-level feature representation by cascading and aggregating different levels of spatial–spectral information, so that the model can better understand the local semantic information in the image.

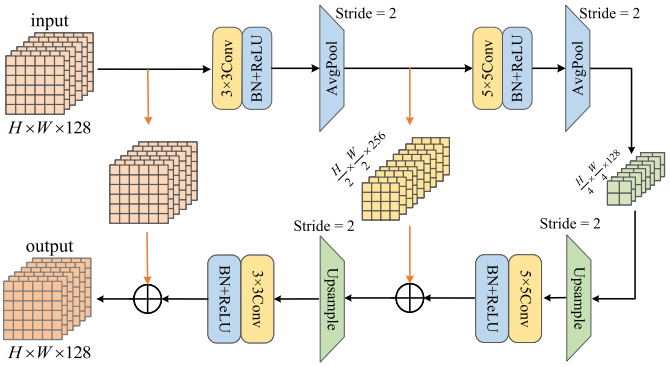


Fig. 3. Structure of the proposed UsU.

Third, we integrate superpixel priors with pixel-level features to generate a superpixel sequence, from which the Transformer encoder extracts more representative global contextual information. Finally, the labels of each training set are predicted with a softmax function by fusion of the global and local features. We denote HSI as $X \in \mathbb{R}^{H\times W\times C}$, where H, W, and C represent the height, width, and number of spectral bands of the HSI, respectively.

A. Noise Suppression Module

1) U-Shaped Unit: To hierarchically extract the spatial and spectral features of HSI while mitigating the impact of noise, we designed the UsU, as shown in Fig. 3. The design of the UsU draws inspiration from the U-Net network architecture [23], which aims to capture features with varying resolutions and semantics through

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

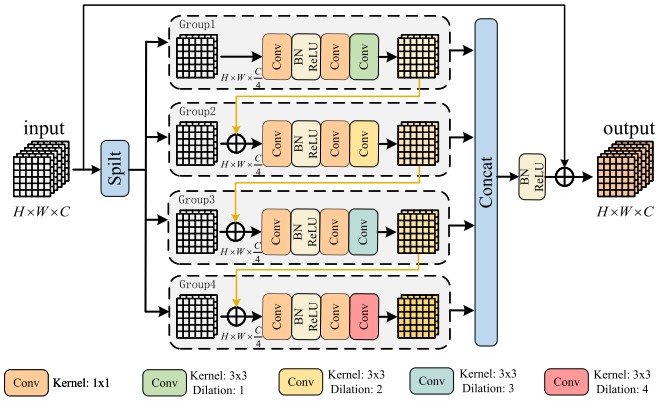


Fig. 4. Structure of the proposed FCU.

downsampling. Subsequently, these features are integrated into a unified feature space via cross-layer fusion and upsampling operations. This multilayer feature extraction and fusion facilitate comprehensive capturing and representation of local details in HSI, thereby enhancing accuracy and robustness in classification and analysis tasks. Moreover, the UsU effectively attenuates perturbations embedded in feature maps, further strengthening the model against noise and variation.

First, before entering the UsU, the input X is subjected to a 1×1 convolution operation to eliminate redundant information from the original HSI data. To make it easier to understand, we denote $f^{k\times k}(\cdot)$ as an operation consisting of the structure $\{\text{Con}v_{k\times k}, \text{BN}, \text{ReLU}\}$, where BN is batch normalization and ReLU is rectified linear unit. The input $\text{X}_{\text{input}} = f^{1\times 1}(X)$ has dimensions $H \times W \times 128$. Second, the specific process of UsU is as follows:

$$\begin{aligned} X_1 &= P_{\text{avg}}(f^{3\times 3}(X_{\text{input}})) \\ X_2 &= P_{\text{avg}}(f^{5\times 5}(X_1)) \\ X_3 &= f^{5\times 5}(\text{Up}(X_2)) \\ X_4 &= f^{3\times 3}(\text{Up}(X_1 + X_3)) \\ X_{\text{UsU}} &= X_{\text{input}} + X_4 \end{aligned} \tag{5}$$

where $P_{\text{avg}}(\cdot)$ denotes the 2×2 average pooling operation and $\text{Up}(\cdot)$ denotes the upsampling operation. After the UsU, the dimension of the output features is consistent with the input.

2) Feature Cascade Unit: Inspired by the concept of group convolution in deep CNN [29], [30], [31], [32], we propose a FCU, as shown in Fig. 4. This module effectively achieves feature diversification through the use of group convolution and dilation convolution. Group convolution facilitates the diversification of features across different groups, while dilation convolution enhances the model's ability to capture spectral and spatial features in HSI data comprehensively. Specifically, dilation convolution is applied with varying dilation rates across different groups, enabling the capture of multilevel features with different receptive fields.

Specifically, we divide the feature maps obtained by in the UsU into four groups equally in the channel dimension. Each group of feature maps undergoes the operations of convolution, normalization, and activation function to obtain the corresponding four groups of features. The feature maps represented by $\tilde{X}'_i$ are computed as follows:

$$\tilde{X}'_i = \begin{cases} \text{ConvBlock}_i(\tilde{X}_i), & i = 1 \\ \text{ConvBlock}_i(\tilde{X}'_{i-1} + \tilde{X}_i), & i > 1 \end{cases} \tag{6}$$

where $\tilde{X}_i$ denotes the ith split of X_{UsU} with dimension $H \times W \times (C/4)$ and $\text{ConvBlock}_i(\cdot)$ denotes the ith ConvBlock group operation. The ConvBlock_i is composed of $\{\text{Con}v_{1\times 1}, \text{BN}, \text{ReLU}, \text{Con}v_{1\times 1}, \text{Con}v_{3\times 3,d=i}\}$.

To facilitate the exchange of information between channels, we use residual connections to add the output of the previous group to the input of the next group. This method enhances the hierarchy of input features for each group and further enhances the diversity of output features.

For the design of the ConvBlock for each group, we referred to the structure with higher robust generalization proposed by Wang et al. [33] and made some modifications. Notably, one of the pivotal modifications involves the utilization of dilated convolutions with different expansion rates. By applying different dilation rates between different groups, multiscale feature extraction and effective information capture are achieved [34], [35], [36], [37]. The ability of the network to identify and model features of varying scale and complexity is further enhanced by this strategy.

Subsequently, we merge the outputs from these four processed channel groups to construct a feature map with the same dimension as the input. This integrated feature map is further processed by normalization and activation function to finally obtain a multiscale feature representation. Formally, this operation can be formulated as follows:

$$\begin{aligned} \tilde{X} &= \text{Concat}(\tilde{X}'_i), \quad 1 \le i \le 4 \\ X_{\text{FCU}} &= X_{\text{UsU}} + \text{ReLU}(\text{BN}(\tilde{X})). \end{aligned} \tag{7}$$

B. Superpixel Sequence Construction

The ability of the Transformer to extract global contextual information is limited by traditional patch-based classification methods. The use of fixed-size patch tokens can also adversely impact the accurate classification of numerous boundary pixels across different categories. To overcome these challenges and enable the Transformer to acquire a comprehensive global perspective during feature extraction, we propose partitioning the whole pixel feature map into ordered sequences of superpixels, so that the Transformer is able to learn useful global contextual information by selectively focusing on different superpixel sequences.

However, due to the varying size of the superpixels and the number of pixels they contain, it is difficult to integrate the transformation between pixels and superpixels into the network. Building on previous research [19], [37], we apply linear discriminant analysis (LDA) for dimensionality reduction preprocessing of HSI and then use an SLIC algorithm to flexibly segment the original HSI into superpixel map. This segmentation ensures a one-to-one correspondence between the raw HSI and superpixel map. We establish this correspondence by generating an association matrix that serves as a bridge for features to mutually transform between pixels

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

and superpixels. The association matrix is denoted as $U \in \mathbb{R}^{HW\times N}$, $N = \lceil (H \times W)/\lambda \rceil$, where N is the number of superpixels and λ is the segmentation parameter that controls the size of the superpixels

$$U_{i,j} = \begin{cases} 1, & \text{if } \bar{X}_i \in S_j \\ 0, & \text{otherwise} \end{cases} \quad \bar{X} = \text{Flatten}(X) \tag{8}$$

where $\bar{X}_i$ denotes the ith pixel in $\bar{X}$, S_j denotes the jth superpixel, and $U_{i,j}$ records the position of each pixel in the superpixel in the original HSI. Therefore, we are able to utilize the association matrix U to define the projection process between the pixel feature vectors and the superpixel sequence. Specifically, let $F \in \mathbb{R}^{H\times W\times 128}$ be the output feature map of the FCU module, so the superpixel sequence S can be written as follows:

$$S = \tilde{U}^T \text{Flatten}(F) \tag{9}$$

where $\tilde{U}$ denotes the normalized U by column. Then, the superpixel sequence is fed into Transformer encoder to capture the global spatial contextual relationships within the HSI.

C. Transformer Encoder

Specifically, the Transformer encoder primarily comprises alternating multihead SA (MHSA) and feedforward (FF) layer. Layer normalization (LN) and residual connection are applied after each block.

1) Layer Normalization: For both the Transformer and its variants, LN is an important normalization method, which is widely used in each multihead attention layer and FF layer. LN normalizes each feature dimension individually for each sample and serves several important purposes, including mitigating gradient vanishing, accelerating convergence, and improving both network performance and stability.

2) Multihead SA: The layer normalized superpixel sequence S is passed through three trainable and shared transformation matrices W^Q, W^K, and W^V to obtain the corresponding query (Q), key (K), and value (V). Subsequently, a dot product operation is applied to compute the queries with all keys. Next, the softmax function is used to compute the attention matrix on values. Finally, dot product operation is utilized to compute the global attention score matrix to values. The calculation of the attention score for SA can be summarized as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{r_K}}\right)V \tag{10}$$

where r_K represents the dimension of the query and key, and multiple independent SA mechanisms are combined to form an SAMHSA module. The formulation for MHSA is provided by the following equation:

$$\begin{aligned} &\text{MHSA}(Q, K, V) = \text{Concat}(\text{SA}_1, \text{SA}_2, \ldots, \text{SA}_\text{h})W^O \\ &\text{SA}_\text{i} = \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right), i = 1, \cdots, h \end{aligned} \tag{11}$$

where h is the number of heads and W^O is the learnable parameter matrix.

3) FF Layer: The FF layer serves as a subsequent step in the transformation of features acquired through the MHSA mechanism. It consists of two fully connected layers, and the parameters of these layers can be shared among all output vectors. Gaussian error linear unit (GELU) is used for all the intermediate nonlinear activation functions. Its equations are defined as follows:

$$\text{FF(x)} = \text{GELU}(xW_1 + b_1)W_2 + \text{b}_2 \tag{12}$$

D. Feature Fusion and Classification

After obtaining the learned features from the Transformer encoder, the shape of the superpixel features needs to be reshaped into the shape of the pixel-level features to complete the pixel-level classification task. This transformation process is as follows:

$$F' = US' \tag{13}$$

where F' denotes the pixel-level features recovered from the superpixel space and S' denotes the superpixel-level features learned from the Transformer encoder. Then, these two different level features are fused with the fusion process

$$\hat{F} = \text{concat}(F'; F) \tag{14}$$

where $\hat{F}$ represents the final fused features. Finally, to train the network, the loss function can be represented as a cross-entropy loss function to optimize the learning process. It can be written as follows:

$$L(\hat{P}, P) = -\sum_{i=1}^{T}\sum_{c=1}^{C} T^{-1} P_c^i \log\left(\hat{P}_c^i\right) \tag{15}$$

where P represents the true label of the sample, $\hat{P}$ represents the predicted label, T represents the number of all training samples, and C represents the number of categories in the training samples.

IV. Experimental Results

In this section, we use the Indian Pines (IP), Pavia University (PU), and Salinas Valley (SV) datasets to validate the effectiveness of the proposed method. The detailed category information of three datasets is shown in Table I.

A. Dataset Description

1) IP Dataset: The IP dataset was captured over an agricultural test site located in the IP region. This HSI comprises 220 bands and has the dimensions of 145 × 145. After excluding bands with interference, a total of 200 bands containing 10 249 samples encompassing 16 terrain classes were chosen for research and analysis.

2) PU Dataset: The PU dataset was acquired over the University of Pavia area. This HSI encompasses 115 bands and has the dimensions of 610 × 340. After excluding bands affected by interference, a total of 103 bands were retained for research purposes. This dataset comprises 42 776 samples across nine terrain classes with labeled annotations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CATEGORY INFORMATION OF THE IP, PU, AND SV DATASETS

ID	Indian Pines				Pavia University				Salinas Valley			
	Class name	Train	Test	Total	Class name	Train	Test	Total	Class name	Train	Test	Total
C1	Alfalfa	20	26	46	Asphalt	100	6531	6631	Weeds_1	100	1909	2009
C2	Corn-N	80	1348	1428	Meadows	100	18549	18649	Weeds_2	100	3626	3726
C3	Corn-P	80	750	830	Gravel	100	1999	2099	Fallow	100	1876	1976
C4	Corn	80	157	237	Trees	100	2964	3064	Fallow_P	100	1294	1394
C5	Grass-M	80	403	483	Metal sheets	100	1245	1345	Fallow_S	100	2578	2678
C6	Grass-T	80	650	730	Bare soil	100	4929	5029	Stubble	100	3859	3959
C7	Grass-P	15	13	28	Bitumen	100	1230	1330	Celery	100	3479	3579
C8	Hay-W	80	398	478	Bricks	100	3582	3682	Grapes	100	11171	11271
C9	Oats	10	10	20	Shadows	100	847	947	Soil	100	6103	6203
C10	Soybeans-N	80	892	972					Corn	100	3178	3278
C11	Soybeans-M	80	2375	2455					Lerruce_4wk	100	968	1068
C12	Soybeans-C	80	513	593					Lerruce_5wk	100	1827	1927
C13	Wheat	80	125	205					Lerruce_6wk	100	816	916
C14	Woods	80	1185	1265					Lerruce_7wk	100	970	1070
C15	Buildings	50	336	386					Vinyard_U	100	7168	7268
C16	Stone	50	43	93					Vinyard_T	100	1707	1807
Total		1025	9224	10249		900	41876	42776		1600	52529	54129

3) SV Dataset: The dataset was collected over the SV. This HSI consists of 224 bands and has the dimensions of 512 × 217. Following the exclusion of bands affected by interference, a total of 204 bands were retained for experimentation. This dataset comprises 54 129 samples representing 16 different crop categories.

B. Experimental Setting

The experiments are organized into four main parts. The first part evaluates the adversarial robustness of various HSI classification methods. To provide a comprehensive assessment, we generated adversarial examples using the FGSM algorithm with a unitized ℓ_∞ norm (2), wherein ε is fixed at 0.06 and targeted category y_{target} in the attack is set as class 1. The second part examines the impact of different perturbation sizes and diverse attack methods, including PGD and CW attacks, on the classification methods. The third part is ablation experiment on the proposed method to evaluate the impact of the proposed module on the performance of the adversarial defense. Finally, the complexity of the networks is analyzed.

In this article, the performance of different classification methods is evaluated using OA, average accuracy (AA), and kappa coefficient (kappa). Each experiment is repeated ten times, with results analyzed based on the mean and standard deviation. The experiments were conducted using the PyTorch framework with the Adam optimizer [38], where the learning rate was set to 0.001 and the number of epochs to 200. For superpixel segmentation scale, the parameter λ was set to 200 for the PU and SV datasets and 50 for the IP dataset. All experiments were performed on a computer with an Intel[1] Core[2] i9-10900K CPU (3.70 GHz), NVIDIA GeForce RTX 3060 GPU, and 32 GB of RAM.

[1]Registered trademark.
[2]Trademarked.

C. Classification Results

In this section, we compare the proposed method with several classical and recent related methods, including 2D-CNN [39], SSFCN [40], pResNet [41], ViT [42], GTFN [4], GAHT [43], SACNet [16], CEGCN [44], and WFCG [45]. Detailed information regarding the network frameworks and optimal parameter settings for these methods can be found in references. The classification results for IP, PU, and SV datasets are presented in Tables II–IV. Based on the classification performance of these methods on adversarial examples across the three datasets, it is clear that many HSI classification methods experience a significant decline in accuracy when faced with adversarial examples. These methods fail to achieve accuracy levels similar to those of clean examples and exhibit substantial standard deviation in classification results. These findings indicate that the intrinsic resistance of these methods is poor, and they are easily interfered by the adversarial attacks. It is worth noting that when comparing pResNet and GAHT, two methods that classify based on patches, Transformer (such as GAHT), do not consistently outperform CNN in terms of robustness. GAHT demonstrates superior classification results on the PU and SV datasets. However, its OA on the IP dataset is notably lower at only 48%, compared with pResNet. This discrepancy may be attributed to the IP dataset having a more pronounced category imbalance issue, where certain categories have relatively fewer samples. Consequently, GAHT may be constrained by the limited number of samples during training, making it challenging to learn sufficiently robust feature representations. In contrast, the PU and SV datasets exhibit a more balanced distribution of categories, which facilitates GAHT in learning effective feature representations from a larger number of samples.

SACNet, CEGCN, WFCG, and the proposed APNet show excellent performance when confronted with adversarial examples generated by FGSM, which shows the importance of whether the network is able to pay attention to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

TABLE II

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	2D-CNN [38]	SSFCN [39]	pResNet [40]	ViT [41]	GAHT [42]	GTFN [4]	SACNet [16]	CEGCN [43]	WFCG [44]	Ours
1	33.93±15.5	93.75±1.86	90.22±9.25	27.81±9.78	8.61±3.52	**99.61±1.15**	98.56±1.86	74.98±12.30	53.46±19.74	89.9±13.13
2	16.60±10.13	76.77±11.78	51.60±6.25	76.16±11.03	37.14±9.49	75.28±7.82	79.71±3.27	94.33±3.22	81.86±13.87	**97.26±1.60**
3	20.51±29.9	64.78±18.37	93.14±4.04	55.84±15.02	37.24±9.80	78.91±11.27	85.00±9.62	**97.91±1.26**	88.24±15.06	95.26±3.22
4	40.37±29.22	90.29±15.55	91.16±7.76	47.55±21.43	50.72±15.79	93.63±2.86	**96.74±3.16**	91.14±5.26	81.76±11.62	93.88±4.03
5	30.41±7.96	84.40±16.65	77.24±16.79	76.89±9.37	57.58±10.41	93.08±3.27	90.32±4.13	**99.23±0.65**	98.75±2.02	98.07±0.88
6	60.86±25.81	85.98±8.42	94.85±2.97	86.17±6.35	95.51±3.31	94.26±4.70	92.00±4.83	99.75±0.22	**99.87±0.16**	98.20±1.43
7	30.99±18.24	98.08±3.33	54.56±21.64	26.34±10.98	28.28±31.26	94.62±6.92	**99.04±2.54**	95.91±8.30	82.58±19.71	92.49±14.79
8	98.22±1.51	97.36±1.52	98.87±1.50	93.58±5.27	50.14±29.24	98.92±1.25	95.82±1.24	**99.63±0.45**	99.35±0.90	99.63±0.40
9	18.18±10.31	98.75±3.31	73.17±21.47	12.61±7.86	20.03±10.04	86.00±13.56	**100.00±00.00**	98.75±3.31	88.88±9.28	82.70±14.48
10	8.75±4.74	57.47±25.11	82.26±8.74	54.02±18.92	43.22±4.30	67.67±12.61	90.11±3.79	90.84±2.89	84.71±5.57	**92.95±3.07**
11	31.50±8.03	47.88±23.02	97.60±1.28	89.20±6.16	65.21±11.54	59.68±7.46	76.02±11.56	97.08±1.64	96.03±3.33	**99.12±0.90**
12	27.22±26.73	49.25±7.41	76.22±2.32	53.12±9.06	18.70±10.34	86.36±3.96	85.41±4.49	**97.71±0.92**	91.25±8.33	98.13±1.26
13	92.38±5.98	93.80±3.46	95.11±2.32	66.11±13.57	75.65±14.05	**99.68±0.53**	92.8±14.83	98.74±1.66	99.51±0.78	98.47±2.33
14	70.98±15.03	69.77±21.72	93.92±6.52	90.46±7.32	93.00±5.27	92.34±6.88	98.11±0.88	99.64±0.36	**99.79±0.24**	99.42±0.52
15	25.29±17.76	70.42±9.44	82.20±9.03	58.12±10.78	64.67±7.24	84.82±5.07	91.52±12.17	**98.55±1.42**	97.69±1.54	97.41±2.40
16	42.42±12.08	93.31±4.85	79.00±6.28	32.52±9.99	74.10±14.60	**100.00±00.00**	97.97±4.57	89.80±4.48	94.30±4.19	95.91±3.05
OA(%)	29.09±4.27	66.34±10.66	78.22±3.72	65.49±5.94	48.09±3.22	77.98±2.35	85.98±2.55	96.66±0.65	90.79±5.21	**97.50±0.64**
AA(%)	46.44±4.74	79.50±4.89	87.96±2.40	80.13±3.36	57.62±3.55	65.71±2.32	91.82±1.79	95.25±1.37	89.88±3.33	**95.55±1.98**
Kappa×100	19.63±4.51	62.09±11.48	75.22±4.17	61.48±6.23	41.89±3.19	75.29±2.57	84.00±2.84	96.15±0.75	89.47±5.88	**97.12±0.73**

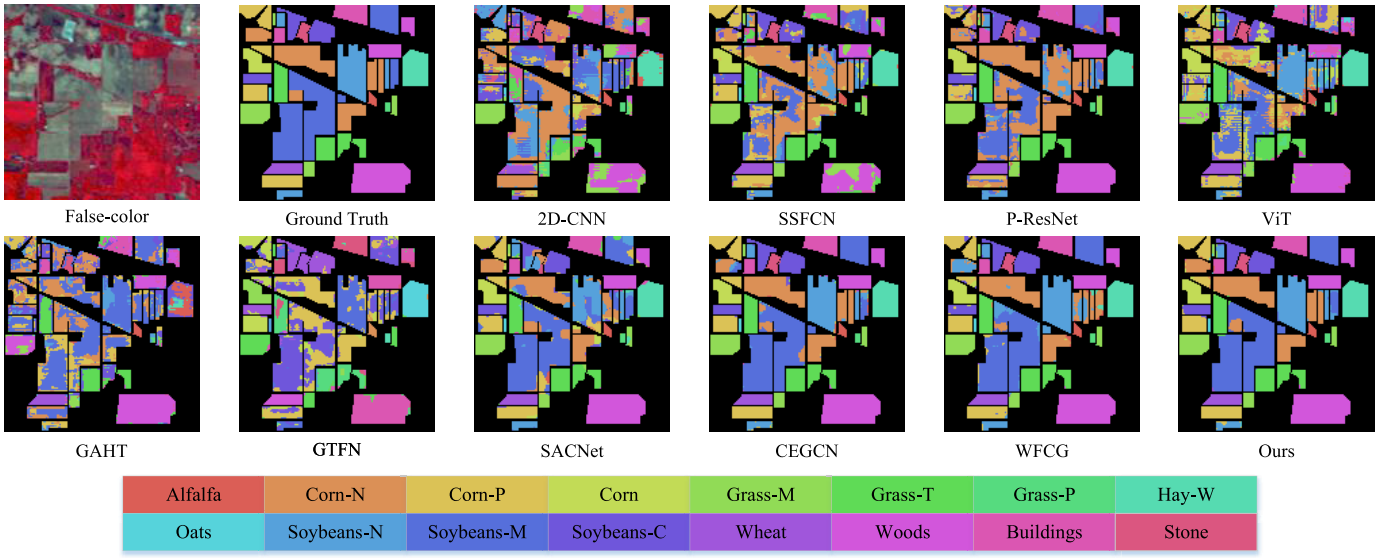


Fig. 5. False-color image, ground truth, and classification maps obtained by different methods on the IP dataset.

spatial context information of HSI. Specifically, SACNet leverages attention mechanisms and global context encoding to acquire pixel-level contextual information, leading to commendable performance. CEGCN combines the power of CNN and GCN to extract both pixel-level and superpixel-level information, surpassing SACNet in terms of accuracy. Unlike CEGCN, WFCG employs graph attention networks to extract superpixel-level features. In contrast, our proposed APNet utilizes the UsU and FCU to obtain pixel-level multiscale semantic information and captures the global spatial contextual information at the superpixel-level using Transformer encoder. By fusing pixel-level and superpixel-level global spatial dependencies, we significantly improve the intrinsic immunity of the network. It consistently achieves an OA of more than 97% across all three datasets, accompanied by lower standard deviations compared with the other two superior-performing networks. This proves that our method is more stable and more capable against adversarial examples, far better than current state-of-the-art DL methods.

To assess the impact on the classification results of the above methods in an intuitive manner, we further provide the classification visualization maps of these methods under the three datasets, as shown in Figs. 5–7. For instance, when examining the classification map of the SV dataset, it becomes apparent that several regions, such as Grapes and Vinyard_U in the top-left corner, are not effectively classified by most classification methods when subjected to adversarial attacks. Furthermore, the soil region in the bottom-left corner is frequently misclassified by many methods as Corn or Fallow. In contrast, our proposed APNet exhibits superior adversarial robustness, and the classification map is closest to the ground truth, which can be observed similarly in IP and PU datasets.

D. Influence of Different Perturbations and Attack Methods

In this section, we employ the FGSM algorithm (2) with a unitized ℓ_∞ norm to generate adversarial examples with increasing attack strengths ranging from 0.01 to 0.1. We then assess the classification performance of different methods when confronted with adversarial examples of varying attack strengths. These series of adversarial examples are generated and evaluated to provide deeper insights into the robustness

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PU DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	2D-CNN [38]	SSFCN [39]	pResNet [40]	ViT [41]	GAHT [42]	GTFN [4]	SACNet [16]	CEGCN [43]	WFCG [44]	Ours
1	84.39±11.58	98.00±0.95	78.99±4.75	77.56±8.54	87.84±3.88	35.33±19.23	85.84±2.67	96.30±2.03	94.17±3.72	**99.18±0.50**
2	44.86±13.61	17.75±26.25	52.17±9.97	90.86±4.09	97.51±0.87	41.46±9.77	87.10±4.84	**99.83±0.14**	99.76±0.38	99.78±0.17
3	6.12±3.91	1.04±0.83	23.18±5.59	55.50±9.27	66.05±6.28	69.96±8.76	84.53±5.70	85.78±16.63	99.22±1.57	**97.93±2.7**
4	88.28±7.05	96.69±0.75	80.30±16.47	50.45±12.43	93.56±2.35	79.38±10.59	93.62±1.43	**99.45±0.30**	98.69±0.82	97.45±5.25
5	99.44±0.49	98.44±0.88	97.55±1.44	97.07±2.27	99.82±0.30	98.97±1.10	99.09±0.68	**99.99±0.03**	99.95±0.04	99.96±0.04
6	3.60±1.62	51.34±26.08	10.18±4.29	31.96±6.15	70.53±11.42	40.42±12.17	83.28±6.25	**98.86±0.97**	96.87±5.10	94.64±4.85
7	8.90±8.94	37.81±16.09	24.35±25.96	46.02±9.69	91.4±5.20	66.09±16.70	81.22±17.93	97.91±5.26	99.66±0.87	**99.97±0.06**
8	69.13±4.42	75.42±8.61	79.54±6.37	59.39±4.54	91.9±3.36	30.40±12.69	93.22±1.35	93.19±5.06	96.69±4.49	**99.33±0.85**
9	94.05±2.79	97.90±0.52	94.54±2.59	46.21±10.41	99.17±0.70	90.46±9.19	97.31±1.40	**99.97±0.08**	99.96±0.08	98.79±0.89
OA(%)	40.53±6.62	48.55±9.90	45.98±3.51	56.55±4.46	88.67±3.33	46.90±5.47	88.49±1.20	97.70±1.26	98.06±1.07	**98.69±0.76**
AA(%)	53.33±4.07	63.82±3.28	58.61±1.89	71.47±2.65	86.59±3.06	44.13±3.16	89.85±1.94	96.81±2.41	98.33±0.74	**98.58±0.76**
Kappa×100	28.21±6.68	40.69±9.48	35.16±3.77	48.14±4.38	85.17±4.21	37.86±5.52	84.89±1.46	96.93±1.67	97.41±1.42	**98.25±1.00**

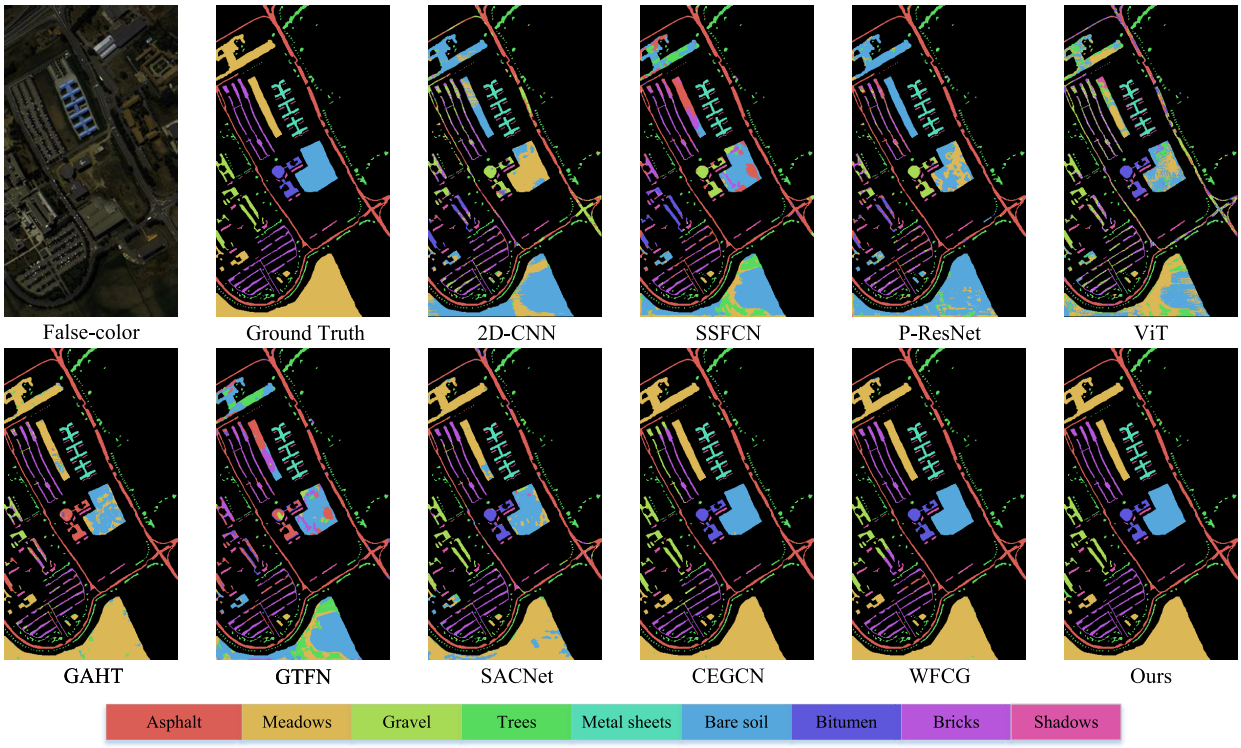


Fig. 6. False-color image, ground truth, and classification maps obtained by different methods on the PU dataset.

TABLE IV

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE SV DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	2D-CNN [38]	SSFCN [39]	pResNet [40]	ViT [41]	GAHT [42]	GTFN [4]	SACNet [16]	CEGCN [43]	WFCG [44]	Ours
1	34.91±10.79	99.82±0.11	98.75±3.31	95.18±5.03	53.12±2.26	99.92±0.12	98.67±1.43	99.47±1.24	95.42±12.13	**100.00±00.00**
2	59.77±21.60	56.88±17.25	59.74±8.67	73.99±16.46	99.99±0.03	94.73±5.48	87.4±28.99	**100.00±00.00**	**100.00±00.00**	99.99±0.01
3	24.36±29.97	0.52±0.66	5.72±4.63	56.05±23.94	71.53±4.20	91.96±6.21	78.42±20.43	99.91±0.09	**99.94±0.12**	99.78±0.38
4	79.24±14.74	48.31±37.53	86.50±18.55	43.81±8.14	94.59±1.00	92.61±6.48	92.06±5.79	**99.14±0.96**	98.28±1.19	98.02±1.39
5	69.07±17.92	26.30±27.79	33.41±20.03	78.15±10.13	87.41±1.90	92.58±5.77	97.87±1.42	99.82±0.23	**99.88±0.30**	99.58±0.48
6	99.06±1.41	86.78±30.54	99.05±0.84	95.19±4.81	99.95±0.02	96.39±2.87	99.29±0.52	**99.99±0.02**	99.98±0.02	99.96±0.05
7	94.50±5.47	23.85±33.46	90.45±11.60	48.48±8.68	96.46±3.51	96.56±4.39	93.66±4.51	**99.99±0.02**	99.95±0.06	99.95±0.08
8	42.70±16.41	55.69±38.90	28.91±2.29	69.39±13.97	77.89±5.04	65.29±5.44	77.79±6.32	89.83±5.93	94.88±5.67	**98.32±1.72**
9	79.38±23.54	0.01±0.01	67.52±16.01	99.59±0.70	73.94±4.56	84.08±8.91	80.94±29.43	**99.98±0.04**	99.92±0.14	99.90±0.18
10	79.77±16.21	72.22±21.46	26.84±3.49	83.57±5.75	49.92±13.39	92.49±5.39	97.77±0.96	98.60±1.46	**98.71±1.21**	96.92±2.68
11	57.87±21.5	55.01±31.47	17.76±14.32	34.14±7.25	31.14±13.41	98.08±1.40	98.09±2.12	99.78±0.44	**99.99±0.03**	98.42±3.75
12	57.72±22.24	5.55±7.85	15.59±15.92	24.40±7.90	99.75±0.10	91.96±5.03	94.49±12.03	99.85±0.41	**100.00±00.00**	99.98±0.04
13	36.01±10.70	76.00±32.92	85.13±17.77	51.13±12.19	99.90±0.13	93.68±7.35	99.51±0.60	99.99±0.04	99.77±0.60	**100.00±00.00**
14	49.93±9.13	45.44±39.19	83.72±10.25	46.15±27.57	92.6±4.05	94.94±3.45	98.67±0.65	**99.66±0.55**	99.42±0.74	98.43±1.78
15	28.41±11.39	23.49±25.57	7.77±4.57	49.05±15.72	41.69±2.81	72.66±7.86	75.03±13.02	82.80±6.32	81.45±12.00	**93.95±3.99**
16	55.37±21.05	28.78±25.04	82.38±16.93	33.63±7.05	99.41±0.40	98.95±1.09	98.87±0.71	99.98±0.04	**100.00±00.00**	99.48±0.83
OA(%)	49.60±8.88	41.59±8.62	40.58±1.80	52.49±3.05	65.72±1.77	84.35±1.81	86.76±4.49	95.31±2.03	95.07±2.29	**98.41±0.84**
AA(%)	54.91±5.13	44.04±6.68	49.24±3.72	61.76±2.94	76.09±2.12	81.51±2.11	91.78±3.81	98.05±0.81	97.97±0.81	**98.92±0.57**
Kappa×100	44.37±9.50	35.83±8.38	34.23±1.99	48.32±3.32	62.28±2.00	82.80±1.97	85.28±4.99	94.76±2.27	94.51±2.53	**98.23±0.93**

of the network, specifically its performance under different levels of adversarial perturbations. The classification results are shown in Fig. 8. From the figure, it is evident that the OA of many methods gradually decreases as the attack intensity increases. This trend shows that larger perturbations lead to reduced network resistance to adversarial examples,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

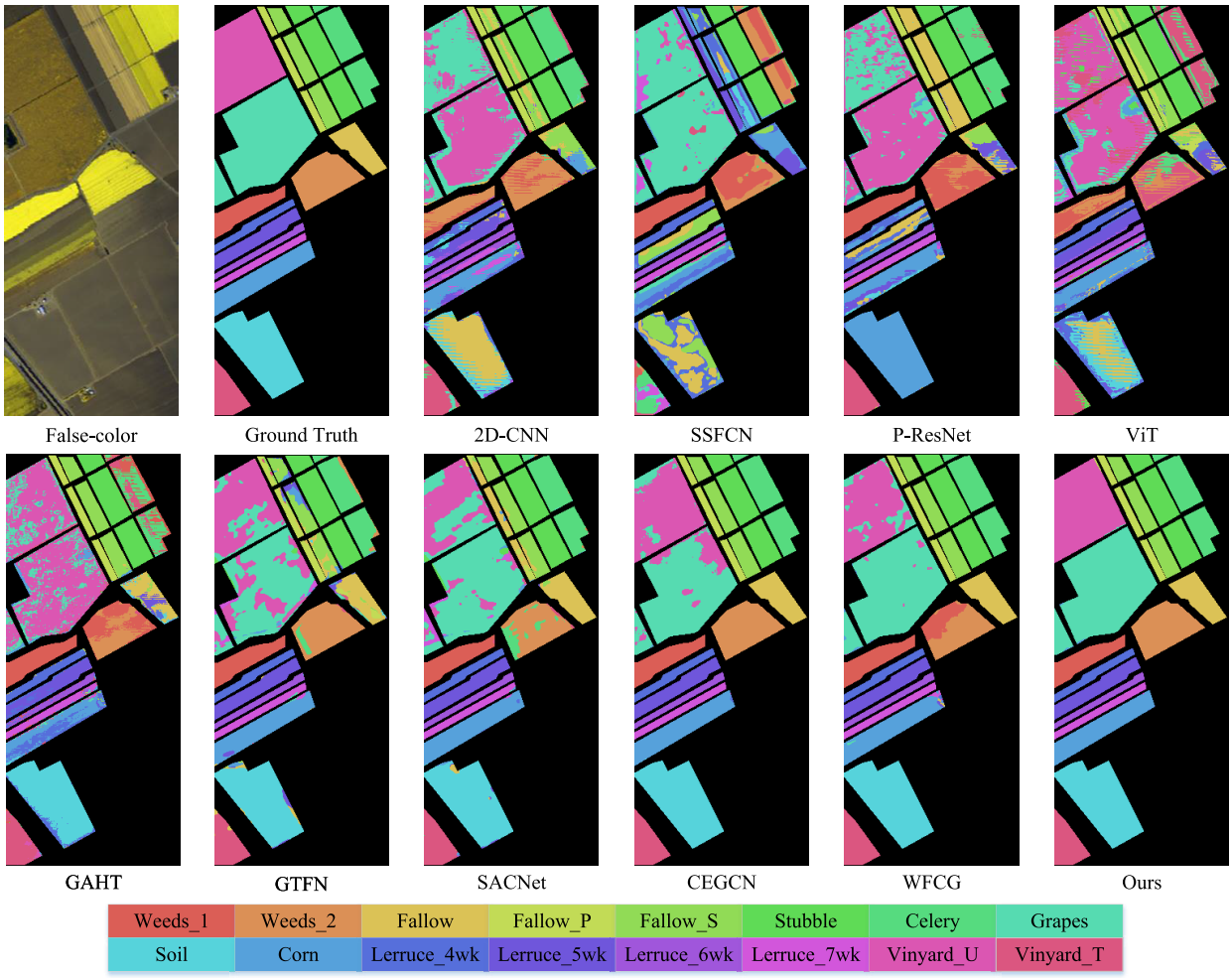


Fig. 7. False-color image, ground truth, and classification maps obtained by different methods on the SV dataset.

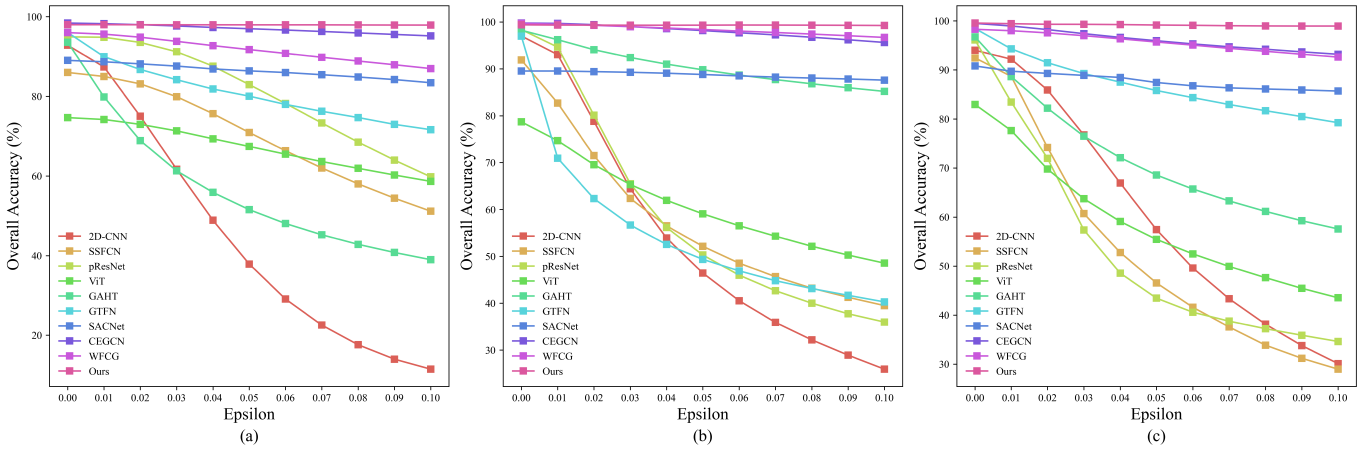


Fig. 8. OAs of different methods on the adversarial test set with varying values of ε. (a) IP. (b) PU. (c) SV.

highlighting the vulnerability of HSI classification methods. In addition, our proposed method consistently exhibits the best classification performance on all three datasets compared with the more robust SACNet, CEGCN, and WFCG. This result emphasizes the effectiveness of establishing global spatial dependencies at the pixel-level and superpixel-level to improve the intrinsic resistance of the network and, thus, enhance its adversarial robustness.

Based on the above experiments, we further compare the performance of different methods under alternative adversarial attacks, specifically the TPGD [26], PGD, and C&W attacks. For these attacks, we use the ℓ_∞ norm to generate adversarial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

TABLE V
OVERALL ACCURACIES OF DIFFERENT METHODS ON THE CLEAN TEST SETS AND ADVERSARIAL TEST SETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	2D-CNN [38]	SSFCN [39]	pResNet [40]	ViT [41]	GAHT [42]	GTFN [4]	SACNet [16]	CEGCN [43]	WFCG [44]	Ours
Indian Pines										
Clean Test Set	92.85	85.99	94.95	74.67	93.60	96.05	89.02	**98.41**	96.01	97.66
FGSM	29.09	66.34	78.22	65.49	48.09	77.98	85.98	96.66	90.79	**97.50**
PGD	20.60	25.46	33.80	42.39	18.54	48.28	70.41	82.06	85.90	**96.88**
TPGD	39.57	75.19	23.22	75.00	35.34	78.24	84.98	59.03	55.71	**88.58**
CW	20.93	26.56	33.63	47.89	22.26	51.41	72.28	52.43	81.36	**96.48**
Pavia University										
Clean Test Set	97.02	91.90	98.38	78.71	98.24	96.99	89.53	**99.77**	99.70	99.01
FGSM	40.53	48.55	45.98	56.55	88.67	46.90	88.49	97.70	98.06	**98.69**
PGD	35.81	25.64	39.92	34.94	62.40	26.81	81.97	79.72	92.53	**98.20**
TPGD	35.12	62.32	19.99	71.98	46.72	51.63	79.09	83.66	68.84	**97.21**
CW	38.10	25.70	40.84	41.10	69.10	32.00	83.59	67.31	88.36	**96.04**
Salinas Valley										
Clean Test Set	94.02	92.45	96.11	82.96	96.75	98.46	90.80	**99.49**	98.29	98.90
FGSM	49.60	41.59	40.58	52.49	65.72	84.35	86.76	95.31	95.07	**98.41**
PGD	33.33	16.46	41.49	37.11	12.24	48.16	69.03	82.78	92.49	**93.87**
TPGD	15.56	74.04	9.66	74.39	30.15	90.66	84.03	70.12	83.99	**94.96**
CW	34.81	19.38	39.59	36.50	18.98	53.31	70.77	76.36	90.23	**94.47**

examples and set the attack perturbation to 0.06, consistent with the FGSM attack, and the number of iterations to be 5. The experimental results presented in Table V demonstrate that the PGD and C&W attacks pose a more harmful to HSI classification methods. Many networks that exhibit high robustness against the FGSM attack are revealed to be more vulnerable when facing the PGD and C&W attacks. For example, consider CEGCN on the PU dataset. The OA for clean examples is 99.77%, which drops by only 2% when confronted with adversarial examples generated by the FGSM attack. However, when confronted with adversarial examples generated by the PGD and C&W attacks, the OA drops drastically by 20% and 32%, respectively. Similar phenomena are observed in other classification methods as well. In contrast, while the accuracy of our proposed method on clean examples may occasionally be lower than that of CEGCN and WFCG, our performance under all four attack methods consistently outperforms that of other classification methods.

E. Ablation Study

In this section, we further evaluate how each module in the proposed APNet contributes to the classification performance on adversarial examples. The adversarial examples are generated using a CW attack with ℓ_∞ norm, where the attack perturbation is set to 0.06 and the number of iterations is 5.

To explore the gain of UsU and FCU on the classification performance of APNet, this article designs ablation experiments on the three datasets to verify the effectiveness of the two modules for HSI classification. The results are shown in Table VI. Specifically, we can conclude that both the UsU and the FCU can significantly improve the robustness of baseline (APNet w/o UsU and FCU) in adversarial example. Taking the SV dataset as an example, the classification accuracy without using UsU and FCU is only 69%. When the UsU is added, the accuracy improves by 21%; with the addition of FCU, the accuracy increases by 16%; and when both UsU and FCU are added, the accuracy improves by 25%. This indicates that the introduction of UsU and FCU is crucial for improving the adversarial robustness of the model in adversarial example.

TABLE VI
OVERALL ACCURACIES OF DIFFERENT MODULES ON DIFFERENT DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Cases	(1)	(2)	(3)	(4)	(5)	(6)
UsU	-	✓	-	-	✓	✓
FCU	-	-	✓	-	-	✓
Self-Attention Block	-	-	-	✓	✓	-
Indian Pines	72.22	93.44	83.15	81.05	85.21	**96.48**
Pavia University	84.15	94.61	92.68	93.55	**97.61**	96.04
Salinas Valley	69.63	90.04	85.38	81.93	89.76	**94.47**

TABLE VII
ABLATION STUDY RESULTS OF PIXEL AND SUPERPIXEL FEATURES ON DIFFERENT DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Cases	(1)	(2)	(3)
Pixel Feature	✓	-	✓
Superpixel Feature	-	✓	✓
Indian Pines	35.86	94.66	**96.48**
Pavia University	54.28	93.76	**96.04**
Salinas Valley	66.55	93.22	**94.47**

To further explore the effectiveness of different scales of dilation convolution in the FCU module for global semantic information extraction, we conducted experiments to evaluate the effect of incorporating the SA block compared with different scales of dilation convolution. The results, presented in Table VI, demonstrate that both the SA block and dilated convolutions can significantly improve adversarial robustness. In particular, using SA block achieves the best performance on the PU dataset, but it does not perform as well as multiscale dilated convolutions on the other two datasets. This suggests that while the SA block can enhance performance in certain scenarios, they may not fully replace the role of multiscale dilated convolutions in the FCU module.

In addition to the network structure, the superpixel-level and pixel-level features in the model have a critical impact on adversarial robustness. The analysis in Table VII indicates that the removal of superpixel-level features has a greater impact on adversarial robustness. Specifically, there was a decrease of 61% and 42% for IP and PU, respectively, when superpixel-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

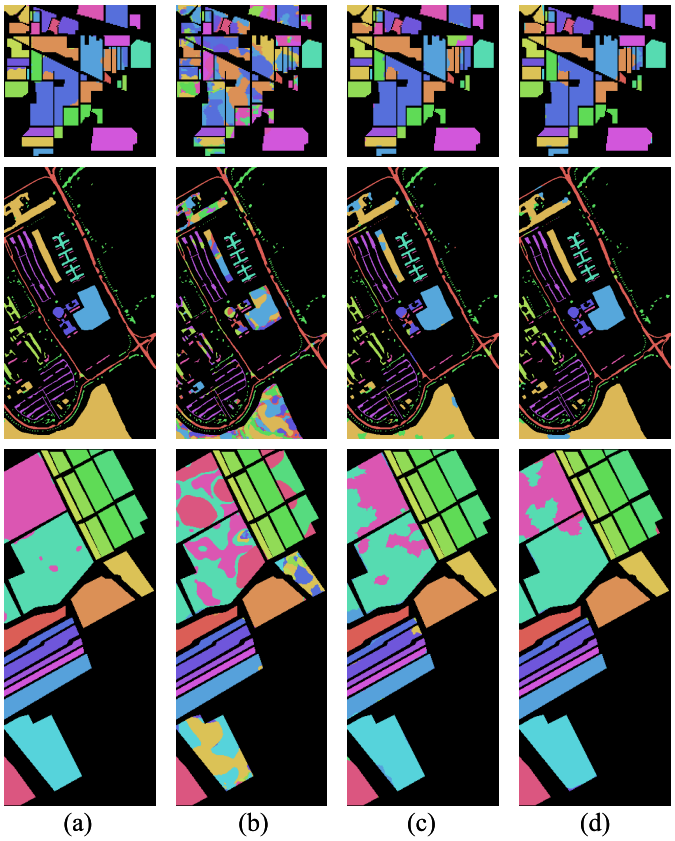


Fig. 9. Ablation visualization of APNet model on three datasets. (a) Classification results of APNet on clean HSI. (b) Classification results of pixel-level feature on adversarial examples. (c) Classification results of superpixel-level feature on adversarial examples. (d) Classification results of APNet on adversarial examples.

level features were removed. This highlights the important role of superpixels in integrating global spatial information, which has a critical impact on improving the adversarial robustness of the model. In contrast, the removal of pixel-level features has a relatively small impact on the adversarial robustness of the model, suggesting that pixel-level features play a secondary role in this context, complementing local detail and texture information.

To visually demonstrate the impact of superpixel-level features and pixel-level features on classification results, we have visualized their classification results. Fig. 9 shows the classification results for the three datasets. We can observe that the classification map generated with superpixel-level features is smoother, but this also leads to some classification errors in small areas. However, after integrating pixel-level features, these small area classification errors are reduced, and the OA and detail performance of the classification results are improved. This further emphasizes the importance of combining pixel-level and superpixel-level information to improve network performance and adversarial robustness.

F. Comparison of Computational Efficiency

In this section, we delve into the complexity analysis of the methods. We present the training and test times of various methods on the three datasets in Table VIII. Among the more robust classification methods (SACNet, CEGCN, and WFCG), the training and testing time of our proposed method is only slightly inferior to that of SACNet. However, it is worth noting that while CEGCN also exhibits good adversarial robustness, its training time is twice that of our method.

TABLE VIII
TRAINING TIME AND TEST TIME OF ALL CLASSIFICATION METHODS ON THE THREE DATASETS (IN SECONDS)

Method		Training Time (s)			Test Time (s)		
		IP	PU	SV	IP	PU	SV
Patch-based Methods	2D-CNN	219.99	51.35	612.21	3.58	21.24	44.14
	pResNet	253.16	185.48	352.91	2.61	10.99	16.57
	ViT	201.85	122.59	343.9	3.68	69.86	21.12
	GAHT	79.49	59.11	123.92	4.2	35.78	19.62
	GTFN	683.26	636.87	1184.77	14.45	67.06	89.35
Images-based Methods	SSFCN	12.24	61.06	35.75	0.002	0.001	0.002
	SACNet	15.86	101.49	55.16	0.003	0.003	0.003
	CEGCN	29.18	271.58	137.82	1.03	4.53	3.8
	WFCG	18.46	306.4	113	0.7	4.17	3.86
	Ours	15.19	126.97	63.42	0.68	4.09	2.54

TABLE IX
FLOPS AND PARAMETERS OF ALL CLASSIFICATION METHODS ON THE THREE DATASETS

Method		FLOPs (M)			Params (M)		
		IP	PU	SV	IP	PU	SV
Patch-based Methods	2D-CNN	15.16	13.75	15.21	0.42	0.4	0.42
	pResNet	30.14	28.45	28.45	1.13	1.11	1.11
	ViT	17.86	15.68	17.95	2.97	2.73	2.98
	GAHT	50.62	45.2	47.37	1.04	0.93	0.97
	GTFN	741.69	726.42	741.69	0.29	0.28	0.29
Images-based Methods	SSFCN	2310	20070	12240	0.11	0.1	0.11
	SACNet	3310	22090	18240	0.21	0.15	0.21
	CEGCN	11.11	21.47	16.74	0.17	0.15	0.17
	WFCG	341.37	795.64	510.94	3.2	3.19	3.2
	Ours	164.01	373.47	240.87	3.79	3.78	3.79

We compared the complexity of different models by calculating the number of floating point operations (FLOPs) and the number of parameters. The results are shown in Table IX. Compared with other networks, our method maintains a moderate FLOPs, but has a larger number of parameters. Nevertheless, APNet still performs excellently in terms of real-time capability and adversarial robustness, with its test time remaining within 4 s. This significant advantage makes APNet more competitive and valuable for industrial applications that require real-time processing and low latency.

V. CONCLUSION

In this article, we introduced a novel antiperturbation network, APNet, specifically designed to defend against adversarial attacks in HSI. APNet incorporates a noise suppression module to extract clear pixel-level features at multiple scales. These features are then combined with superpixel priors to generate robust tokens, which are subsequently fed into Transformer encoder to capture global structural features. By fusing these global structural features with the denoised pixel-level features, APNet significantly enhances the network's intrinsic resistance to adversarial attacks. Extensive experiments on three HSI datasets demonstrate that APNet consistently offers superior adversarial robustness across four different types of adversarial attacks, outperforming the existing SOTA robust classification methods. These results confirm APNet's effectiveness in improving the reliability and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.

security of HSI classification under adversarial conditions. Future work will focus on developing new HSI-specific adversarial defense strategies, further enhancing the network's architecture, and extending its applicability to other domains facing similar challenges.

REFERENCES

[1] V. Kozoderov, T. Kondranin, E. Dmitriev, and V. Kamentsev, "A system for processing hyperspectral imagery: Application to detecting forest species," *Int. J. Remote Sens.*, vol. 35, no. 15, pp. 5926–5945, 2014.

[2] X. Guan, H. Shen, X. Li, W. Gan, and L. Zhang, "Climate control on net primary productivity in the complicated mountainous area: A case study of yunnan, China," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 4637–4648, Dec. 2018.

[3] K. Randolph, J. Wilson, L. Tedesco, L. Li, D. L. Pascual, and E. Soyeux, "Hyperspectral remote sensing of cyanobacteria in turbid productive water using optically active pigments, chlorophyll a and phycocyanin," *Remote Sens. Environ.*, vol. 112, no. 11, pp. 4009–4019, Nov. 2008.

[4] A. Yang, M. Li, Y. Ding, D. Hong, Y. Lv, and Y. He, "GTFN: GCN and transformer fusion network with spatial–spectral features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 6600115.

[5] F. Xu, S. Mei, G. Zhang, N. Wang, and Q. Du, "Bridging CNN and transformer with cross-attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522214.

[6] Q. Yu, W. Wei, D. Li, Z. Pan, C. Li, and D. Hong, "HyperSINet: A synergetic interaction network combined with convolution and transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508118.

[7] Q. Yu, W. Wei, Z. Pan, J. He, S. Wang, and D. Hong, "GPF-Net: Graph-polarized fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519622.

[8] J. Yao, D. Hong, C. Li, and J. Chanussot, "SpectralMamba: Efficient mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.

[9] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[10] C. Szegedy et al., "Intriguing properties of neural networks," 2013, *arXiv:1312.6199*.

[11] K. Eykholt et al., "Robust physical-world attacks on deep learning visual classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1625–1634.

[12] A. Zolfi, M. Kravchik, Y. Elovici, and A. Shabtai, "The translucent patch: A physical and universal attack on object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15232–15241.

[13] W. Czaja, N. Fendley, M. Pekala, C. Ratto, and I.-J. Wang, "Adversarial examples in remote sensing," in *Proc. 26th ACM SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2018, pp. 408–411.

[14] L. Chen, G. Zhu, Q. Li, and H. Li, "Adversarial example in remote sensing image recognition," 2019, *arXiv:1910.13222*.

[15] C. Shi, Y. Dang, L. Fang, Z. Lv, and M. Zhao, "Hyperspectral image classification with adversarial attack," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[16] Y. Xu, B. Du, and L. Zhang, "Self-attention context network: Addressing the threat of adversarial attacks for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 8671–8685, 2021.

[17] Y. Xu, Y. Xu, H. Jiao, Z. Gao, and L. Zhang, "S^3ANet: Spatial–spectral self-attention learning network for defending against adversarial attacks in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512913.

[18] J. Qi, Z. Gong, X. Liu, C. Chen, and P. Zhong, "Masked spatial–spectral autoencoders are excellent hyperspectral defenders," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jan. 1, 2024, doi: 10.1109/TNNLS.2023.3345734.

[19] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[20] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[21] A. Kurakin, I. Goodfellow, and S. Bengio, "Adversarial machine learning at scale," 2016, *arXiv:1611.01236*.

[22] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," 2017, *arXiv:1706.06083*.

[23] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 39–57.

[24] W. Xu, D. Evans, and Y. Qi, "Feature squeezing: Detecting adversarial examples in deep neural networks," 2017, *arXiv:1704.01155*.

[25] D. Meng and H. Chen, "MagNet: A two-pronged defense against adversarial examples," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Oct. 2017, pp. 135–147.

[26] H. Zhang, Y. Yu, J. Jiao, E. Xing, L. El Ghaoui, and M. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 7472–7482.

[27] S. Gu and L. Rigazio, "Towards deep neural network architectures robust to adversarial examples," 2014, *arXiv:1412.5068*.

[28] J. Bradshaw, A. G. de G. Matthews, and Z. Ghahramani, "Adversarial examples, uncertainty, and transfer testing robustness in Gaussian process hybrid deep networks," 2017, *arXiv:1707.02476*.

[29] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.

[30] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1251–1258.

[31] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An extremely efficient convolutional neural network for mobile devices," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6848–6856.

[32] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 1492–1500.

[33] Z. Wang, Y. Bai, Y. Zhou, and C. Xie, "Can CNNs be more robust than transformers?" 2022, *arXiv:2206.03452*.

[34] P. Wang et al., "Understanding convolution for semantic segmentation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Lake Tahoe, NV, USA, Mar. 2018, pp. 1451–1460.

[35] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.

[36] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2017.

[37] R. Hamaguchi, A. Fujita, K. Nemoto, T. Imaizumi, and S. Hikosaka, "Effective use of dilated convolutions for segmenting small object instances in remote sensing imagery," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 1442–1450.

[38] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[39] V. Sharma, A. Diba, T. Tuytelaars, and L. Van Gool, "Hyperspectral CNN for image classification & band selection, with application to face recognition," KU Leuven, ESAT, Leuven, Belgium, Tech. Rep. KUL/ESAT/PSI/1604, 2016.

[40] Y. Xu, B. Du, and L. Zhang, "Beyond the patchwise classification: Spectral–spatial fully convolutional networks for hyperspectral image classification," *IEEE Trans. Big Data*, vol. 6, no. 3, pp. 492–506, Sep. 2020.

[41] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for Spectral–Spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.

[42] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[43] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[44] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel-and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2020.

[45] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.



Lin Zhao (Member, IEEE) received the M.S. degree in computer application technology from the Huazhong University of Science, Wuhan, China, in 2007, and the Ph.D. degree in control science and engineering from Central South University, Changsha, China, in 2017.

As an Associate Professor, he is conducting research programs in optical imaging and hyperspectral image processing. His research interests include image processing, pattern recognition, and modeling and optimal control of the complex industrial process.



Youlin Zhang received the B.S. degree from Taizhou University, Taizhou, China, in 2022. He is currently pursuing the M.S. degree with Hunan Institute of Science and Technology, Yueyang, China.

His research interests include deep learning, adversarial attack, and hyperspectral image classification.



Chengzhong Shi received the M.S. degree from Hunan Institute of Science and Technology, Yueyang, China, in 2024.

His research interests include deep learning, computer vision, and remote sensing image processing.



Minhui Zhao received the B.S. degree from Shanghai Institute of Technology, Shanghai, China, in 2023. She is currently pursuing the M.S. degree with Hunan Institute of Science and Technology, Yueyang, China.

Her research interests include hyperspectral image classification, open-set recognition, and deep learning.



Jianhui Wu received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 2000, and the M.S. and Ph.D. degrees in optical information engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2006 and 2009, respectively.

He was a Visiting Researcher at the Computer Vision Group, University of Nevada, Reno, NV, USA, from August 2015 to August 2016, working on object detection and recognition. He is a Full Professor with the School of Information Science and Technology, Hunan Institute of Science and Technology, Yueyang, China. His main research interests include digital image processing and pattern recognition.



Wen Li received the M.S. degree in automation instruments and devices from the East China University of Technology, Shanghai, China, in 1993, and the Ph.D. degree in thermal power engineering from the University of Shanghai for Science and Technology, Shanghai, in 2004.

He is a Full Professor with the School of Information Science and Technology, Hunan Institute of Science and Technology, Yueyang, China. His main research interests include network controls and digital image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:24:26 UTC from IEEE Xplore. Restrictions apply.